TASK 15: DNS SECURITY BASICS

Cybersecurity Internship | Veda Technology

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1. Objective

To understand how the Domain Name System (DNS) works and identify basic DNS security risks.

2. Introduction

DNS stands for Domain Name System. It translates human-readable domain names, such as example.com, into IP addresses that computers use to communicate. DNS is essential for browsing the internet, but attackers may abuse weaknesses in DNS configuration or responses.

3. How DNS Resolution Works

1. A user enters a domain name in a browser.
2. The device checks its local DNS cache.
3. If the answer is not cached, a DNS resolver looks up the answer, contacting other DNS servers when needed.
4. The resolver returns the IP address to the device.
5. The browser connects to the destination server.

4. Basic DNS Security Risks

Risk	Meaning	Basic Protection
DNS spoofing / cache poisoning	False DNS information may direct users to a malicious destination.	Use trusted resolvers, keep DNS software updated, and use DNSSEC validation where supported.
DNS hijacking	DNS settings or accounts are changed to redirect queries.	Secure registrar/router accounts, change default passwords, and enable MFA.
DNS tunneling	DNS queries are misused to carry hidden data or command traffic.	Monitor unusual DNS patterns and use suitable security monitoring.
Misconfiguration	Incorrect records or overly permissive settings can cause outages or expose information.	Review records, restrict administrative access, and audit changes.
Denial of service (DoS/DDoS)	Large volumes of queries can disrupt DNS availability.	Use resilient DNS providers, rate limiting, and DDoS protection.

5. DNSSEC

DNSSEC stands for Domain Name System Security Extensions. It uses digital signatures to help DNS resolvers verify that DNS data is authentic and has not been altered. DNSSEC helps protect against certain forms of DNS spoofing and cache poisoning, but it does not encrypt DNS queries or prevent every DNS attack.

6. Interview Questions and Answers

Q1. What is DNS?
DNS is the system that translates domain names into IP addresses.

Q2. What is DNS spoofing?
DNS spoofing is an attack in which false DNS information is supplied so that a user may be directed to the wrong or malicious destination.

Q3. What is DNSSEC?
DNSSEC is a set of extensions that uses digital signatures to help verify the authenticity and integrity of DNS data.

7. Basic Security Recommendations

- Keep DNS servers, routers, and devices updated.
- Use trusted DNS resolvers and enable DNSSEC validation where available.
- Protect DNS registrar and administrator accounts with strong unique passwords and MFA.

• Review DNS records and monitor unusual query activity.
• Only test or assess systems when you have permission.

8. Conclusion

DNS makes internet navigation possible by resolving domain names to IP addresses. Understanding DNS resolution, spoofing, and DNSSEC provides a foundation for recognizing and reducing common DNS security risks.

9. References

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